

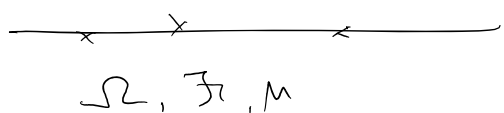
Lebesgue Integration. (real valued f) eg: $\{X(\omega) > -5\}$

defn. ① On $(\Omega, \mathcal{F}, \mu)$ a statement $\lambda(\omega)$ holds a.e almost everywhere if $\mu(\{\omega: \lambda(\omega) \text{ does not hold}\}) = 0$.

if $\mu = P$. can also say

$$P(\{\omega: \lambda(\omega) \text{ holds}\}) = 1$$

also called almost surely (a.s)



defn. ② simple functions. f is simple if

$$f = \sum_{i=1}^k c_i 1_{A_i} \quad \text{for some } A_i \in \mathcal{F}. \quad c_i \in \underline{\mathbb{R}}$$

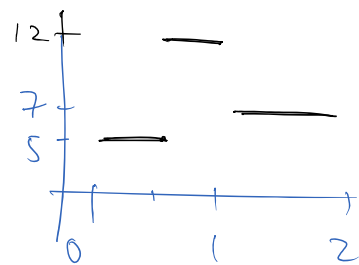
eg: $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$

$$f = 5 1_{(0,1)} + 7 1_{(2,3)}$$

$$g = 5 1_{(0,1)} + 7 1_{(1/2, 2)}$$

$$= 5 1_{(0, 1/2)} + 12 1_{[1/2, 1]} + 7 1_{(1, 2)}$$

$$= 5 1_{(0, 1/3)} + 5 1_{(1/3, 1)} + 7 1_{(1/2, 2)}$$



c_i 's don't need to be distinct, A_i 's don't need to be disjoint.

but there is always one representation with c_i 's distinct and A_i 's disjoint. in that case $A_i = f^{-1}(\{c_i\})$

$$(\Omega, \mathcal{F}, \mu) \xrightarrow{f} (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

$$(\Omega, \mathcal{F}, \mu) \xrightarrow{f} (\mathbb{R}, \mathcal{B}(\mathbb{R}))$$

$$f = 1_A, \quad A \in \mathcal{F}.$$

$$\int f d\mu = 1 \cdot \mu(A)$$

$$f = \sum_{i=1}^k c_i 1_{A_i}$$

$$\int f d\mu = \sum_{i=1}^k c_i \mu(A_i)$$

$$\text{if } \mu = P, \quad \int X dP = E_P(X) = E(X)$$

$$f = X$$

notation of L-int:

$$f : \Omega \rightarrow \mathbb{R}, \quad \langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle \text{ mble.}$$

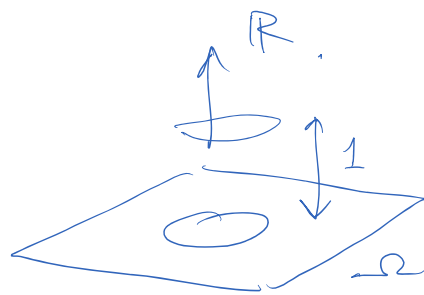
$$\begin{array}{ccc} & \xrightarrow{f} & \\ \hline \Omega, \mathcal{F}, \mu & & \mathbb{R}, \mathcal{B}(\mathbb{R}) \end{array}$$

notation:

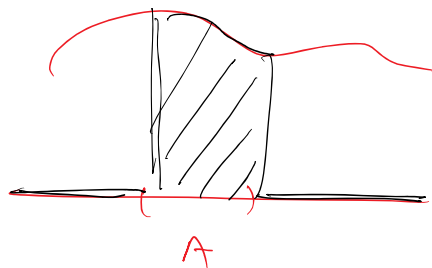
$$\int f d\mu \equiv \int_{\Omega} f(\omega) d\mu(\omega) \equiv \int_{\Omega} f(\omega) \mu(d\omega)$$

$$\text{if } \mu = P, \quad f \equiv X$$

$$\int X dP \equiv \int_{\Omega} X(\omega) dP(\omega) \equiv \int_{\Omega} X(\omega) P(d\omega) \equiv E_P(X) = E(X)$$

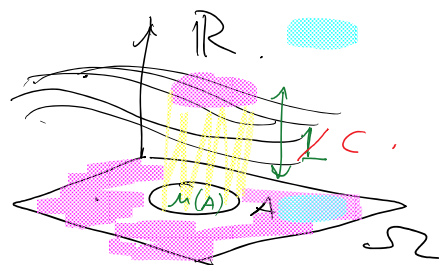


$$\int_A f d\mu \equiv \int f \mathbf{1}_A d\mu$$



$$\textcircled{1} \quad f = c \cdot \mathbf{1}_A$$

$$\int f d\mu = c \cdot \mu(A)$$



$$\textcircled{2} \quad \text{nonneg. simple fns: } f = \sum_{i=1}^K c_i \mathbf{1}_{A_i}, \quad c_i \geq 0, c_i \in \mathbb{R}$$

$$\int f d\mu = \sum_{i=1}^K c_i \mu(A_i)$$

in particular, if f non-neg, simple fns $c_i \geq 0$.

$\textcircled{3}$ If $f \geq 0$ (measurable), can show

$$\textcircled{1} \quad \exists \{f_n\} \text{ nonneg, simple s.t. } f_n \uparrow f$$

Ans: remark 2.3.1 below

$$\text{will define } \int f d\mu \equiv \lim_{n \rightarrow \infty} \int f_n d\mu$$

(lim exists?)

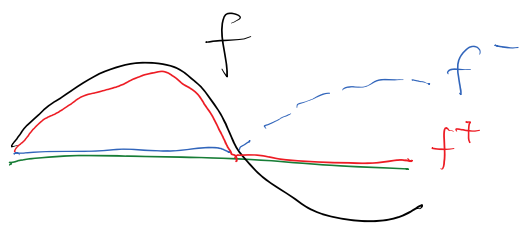
since $\int f_n d\mu \uparrow$ monotonicity (ii) for simple fns.

$\textcircled{2}$ what if $\exists$ two such sequences? (prop. 2.3.2)

basic prop. holds Cor. 2.3.3.

$\textcircled{3}$ it can be $+\infty$.

$\textcircled{4}$ In general f (mble), can split $f = f^+ - f^-$ where $f^+, f^- \geq 0$. Then



$$\int f d\mu \equiv \int f^+ d\mu - \int f^- d\mu$$

↑ already defined

only if one of $\int f^+ d\mu, \int f^- d\mu < \infty$.

We will call f integrable (L-int'ble) only when both $\int f^+ d\mu < \infty$ and $\int f^- d\mu < \infty$.

nonneg simple f

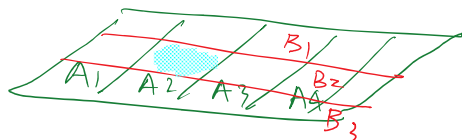
Basic properties (0), (i) - (iii). (p. 2.3.1)

(0) well defined?

$$f = \sum_{i=1}^{k_1} c_i 1_{A_i} = \sum_{j=1}^{k_2} d_j 1_{B_j}$$

$$\sum_{i=1}^{k_1} \sum_{j=1}^{k_2} d_j 1_{A_i \cap B_j}$$

assume A_i 's d_j
 B_j 's d_j .



$$\sum_{i=1}^{k_1} \sum_{j=1}^{k_2} c_i' 1_{A_i \cap B_j}$$

Since f is a function,

$$c_i' = d_j' = e_{ij}$$

$$\text{So } \int f d\mu = \sum_{i=1}^{k_1} \sum_{j=1}^{k_2} e_{ij} \mu(A_i \cap B_j) \rightarrow \text{will be the same}$$

Also note: if $f = \sum_{i=1}^{\textcircled{K}} c_i 1_{A_i}$, $g = \sum_{j=1}^{\textcircled{L}} d_j 1_{B_j}$

Also note: if $f = \sum_{i=1}^n c_i 1_{A_i}$, $g = \sum_{j=1}^m d_j 1_{B_j}$

there is a representation of each of them

$$D_k \approx A_i \cap B_j \quad M = k \times l. \quad f = \sum_{k=1}^M c'_k 1_{D_k}, \quad g = \sum_{k=1}^M d'_k 1_{D_k}$$

(i) $\int (\alpha f + \beta g) d\mu = \alpha \int f d\mu + \beta \int g d\mu.$

Pf. LHS: $\alpha f + \beta g = \alpha \sum_{k=1}^M c'_k 1_{D_k} + \beta \sum_{k=1}^M d'_k 1_{D_k}$
 $= \sum_{k=1}^M (\alpha c'_k + \beta d'_k) 1_{D_k}$

$$\begin{aligned} \int (\alpha f + \beta g) d\mu &= \sum_{k=1}^M (\alpha c'_k + \beta d'_k) \mu(D_k) \\ &= \alpha \sum_{k=1}^M c'_k \mu(D_k) + \beta \sum_{k=1}^M d'_k \mu(D_k) \\ &= \alpha \int f d\mu + \beta \int g d\mu. \end{aligned}$$

(ii) monotonicity $f \geq g$ a.e. (μ)

$$\int f d\mu \geq \int g d\mu.$$

$D = \{\omega : f(\omega) < g(\omega)\}$, $\mu(D) = 0$.

$f = f 1_D + f 1_{D^c}$ by (i) we get -

$$\begin{aligned} \int f d\mu &= \int f 1_D d\mu + \int f 1_{D^c} d\mu \\ &\stackrel{=0}{=} \int_D f d\mu \\ &\quad \left(\equiv \int_D f d\mu \right) \end{aligned}$$

claim $\int f d\mu = 0$.

Let $f = \sum_{i=1}^n c_i 1_{D_i}$

claim $\int_{\mathbb{D}} f d\mu = 0$. , Let $\left. \begin{aligned} f &= \sum_{i=1}^k c_i \underline{1_{D_i}} \\ g &= \sum_{i=1}^k d_i \underline{1_{D_i}} \end{aligned} \right\}$

$$\int_{\mathbb{D}} f d\mu = \int f \underline{1_{\mathbb{D}}} d\mu.$$

$$= \int \left(\sum_{i=1}^k c_i \underline{1_{D_i \cap \mathbb{D}}} \right) d\mu.$$

$$= \sum_{i=1}^k c_i \mu(\overset{+}{\underset{0}{D_i \cap \mathbb{D}}})$$

since $\mu(\mathbb{D}) = 0$.

$$= 0.$$

$$0 \leq \mu(D_i \cap \mathbb{D}) \leq \mu(\mathbb{D}) = 0$$

$$\left\{ \begin{aligned} \int f d\mu &= \int f \underline{1_{\mathbb{D}^c}} d\mu. \\ &= \sum_{i=1}^k c_i \mu(\underline{D_i \cap \mathbb{D}^c}) \end{aligned} \right.$$

$$\left\{ \begin{aligned} f &= \sum_{i=1}^k c_i \underline{1_{D_i}} \\ g &= \sum_{i=1}^k d_i \underline{1_{D_i}} \end{aligned} \right.$$

$$\int g d\mu = \int g \underline{1_{\mathbb{D}^c}} d\mu = \sum_{i=1}^k \underline{d_i \mu(D_i \cap \mathbb{D}^c)} \quad \text{similarly.}$$

note that on $\underline{D_i \cap \mathbb{D}^c}$, $f > g$. (since $\mathbb{D}^c = \{\omega : f(\omega) > g(\omega)\}$)

i.e if $\omega \in \underline{D_i \cap \mathbb{D}^c}$, $f(\omega) > g(\omega)$

" " c_i " d_i on D_i

so $\{ \underline{c_i} > \underline{d_i} \ \forall i \}$

$\Rightarrow \int f d\mu > \int g d\mu$. proved.

(iii) If $f = g$ a.e. (μ) Then $\int f d\mu = \int g d\mu$.

Pf: claim $f = g$ a.e. $\Leftrightarrow \begin{cases} f \geq g \text{ a.e.} \\ \text{also } f \leq g \text{ a.e.} \end{cases}$

$$\{\omega: f(\omega) < g(\omega)\} \subset \{\omega: f(\omega) \neq g(\omega)\}$$

$$\mu(\quad) \leq \mu(\quad) = 0$$

since $f = g$ a.e. (μ)

Since $f \geq g$ a.e. (μ)

$$\int f d\mu \geq \int g d\mu$$

Similarly $\leq$

$$\text{so } \int f d\mu = \int g d\mu \text{ if } f = g \text{ a.e. } (\mu)$$

if A holds a.e.
and $A \Rightarrow B$, Then B holds a.e.
Pf: $B^c \Rightarrow A^c$.
 $\mu(B^c) \leq \mu(A^c) = 0$

$f \geq 0$

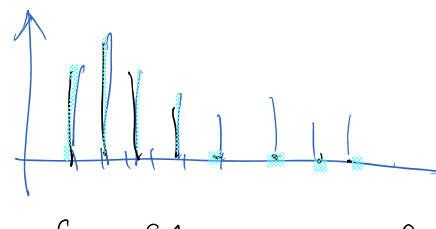
$$\int f d\mu \equiv \lim_{n \rightarrow \infty} \int f_n d\mu, \quad f_n \uparrow f, \quad f_n \geq 0 \text{ simple.}$$

prop. 2.3.2: proof is similar to MCT later.

$f_n \uparrow f$, $f_n \geq 0$, then $\int f_n d\mu \uparrow \int f d\mu$.
don't need to be simple.

ex: $(\Omega, \mathcal{F}, \mu) = (\mathbb{N}, \mathcal{P}(\mathbb{N}), \mu)$
 $f: \mathbb{N} \rightarrow \mathbb{R}^+$

$$\begin{aligned} \int f d\mu &= \lim_{n \rightarrow \infty} \int f_n d\mu \\ &= \sum_{i=1}^{\infty} f(i) \mu(\{i\}) \end{aligned}$$



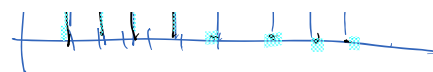
$$= \sum_{i=1}^{\infty} f(i) \mu(\{i\})$$

in particular, if μ is counting measure on $\mathbb{N}$,

$$\int f d\mu = \sum_{i=1}^{\infty} f(i)$$

, i.e. if $f(x) = \frac{1}{x^2}$

$$\int f(x) d\mu(x) = \sum_{n=1}^{\infty} \frac{1}{n^2}$$



$$f_n = f 1_{\{0, n\}} \uparrow f$$

$$\int f_n d\mu = \sum_{i=1}^n f(i) \mu(\{i\})$$

Basic properties $f \geq 0$

$$(i) \int (\alpha f + \beta g) d\mu = \alpha \int f d\mu + \beta \int g d\mu, \quad \alpha \geq 0, \beta \geq 0.$$

proof take $\{f_n\}$ simple $\geq 0 \uparrow f$
 $\{g_n\}$ simple $\geq 0 \uparrow g$.

then $\{\alpha f_n + \beta g_n\}$ is one simple $\geq 0 \uparrow \alpha f + \beta g$.

$$\text{now } \int (\alpha f_n + \beta g_n) d\mu = \alpha \int f_n d\mu + \beta \int g_n d\mu.$$

since these are simple
 (prop 2.3.1)

taking limit ($n \rightarrow \infty$)

on both sides we get the result.

Can verify (ii) & (iii) as well like this.

Fact: if $f \geq 0$ and $\mu(B^c) = 0$ ✓

$$\text{then } \int f d\mu = \int f 1_B d\mu, \quad \int f 1_{\underline{B}^c} d\mu = 0.$$

Pf: can verify as before $\{f_n\} \uparrow f$ etc.

Fact: if $f \geq 0$ and $\mu(B^c) = 0$ ✓

Then $\int f d\mu = \int f 1_B d\mu$, $\int f 1_{\underline{B^c}} d\mu = 0$.

Pf: can verify as before $\{f_n\} \uparrow f$ etc.

alt pf: $f 1_{B^c} = 0$ a.e. (μ)

$$\{\omega: f(\omega) 1_{B^c}(\omega) \neq 0\} \subseteq \{\omega: 1_{B^c}(\omega) \neq 0\} = B^c$$

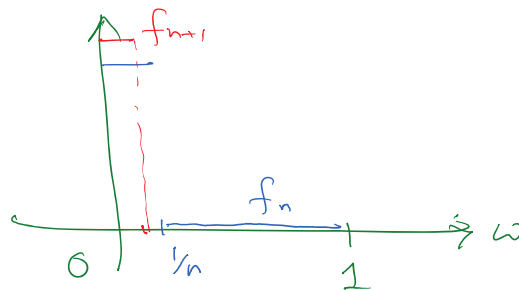
$$\mu(\quad) \leq \mu(B^c) = 0 \leftarrow$$

$$\int f 1_{B^c} d\mu = \int 0 \cdot d\mu = 0.$$

ex: $(\Omega, \mathcal{F}, \mu) = ([0, 1], \mathcal{B}[0, 1], \mu)$ " μ "

$$f_n(\omega) = n 1_{[0, \frac{1}{n}]}(\omega)$$

$$f(\omega) \equiv 0.$$



$$\int f_n d\mu = 1 \quad \forall n \geq 1, \quad \int f d\mu = 0.$$

$$\liminf f_n = f \text{ a.e. } (\mu).$$

But Fatou's Lemma holds!

aside

$$E(\varphi(\theta)) = \int \varphi(\theta) \pi(d\theta)$$

$\uparrow$ $\uparrow$

μ/P w

MCT / Fatou's.

General f , $f \in L_1(\mu)$

$$f = f^+ - f^- \quad = f_1 - f_2 \quad , \quad \begin{matrix} f_1, f_2 \geq 0 \\ f_1, f_2 \in L_1(\mu) \end{matrix}$$

check: $\int f d\mu = \int f_1 d\mu - \int f_2 d\mu.$

Pf:

$$\begin{aligned} (f^+ - f^-) &= f_1 - f_2 \\ \downarrow &\quad (+f_2 + f^-) \qquad\qquad +f_2 + \underline{f^-} \\ f^+ + f_2 &= f_1 + f^- \end{aligned}$$

$$\underbrace{\int f^+ d\mu}_{< \infty} + \underbrace{\int f_2 d\mu}_{< \infty} = \underbrace{\int f_1 d\mu}_{< \infty} + \underbrace{\int f^- d\mu}_{< \infty}.$$

from
Basic
prop.
for non
neg.

$$\int f d\mu \equiv \underbrace{\int f^+ d\mu - \int f^- d\mu}_{\text{proved}} = \int f_1 d\mu - \int f_2 d\mu.$$

ex: combining (null-sets) a.e statements.

$$\left. \begin{array}{l} \mu(D_1^c) = 0 \\ f \geq 0 \\ g \geq 0 \end{array} \right\} \Rightarrow \left. \begin{array}{l} \frac{a \cdot e(\mu)}{a \cdot e(\mu)} D_1 \\ D_2 \end{array} \right\} \Rightarrow \begin{array}{l} D_1 = \{\omega : f(\omega) > 0\} \\ f \geq 0, g \geq 0 \text{ a.e.}(\mu) \\ D = D_1 \cap D_2 \\ 0 \leq \mu(D^c) = \mu(D_1^c \cup D_2^c) \\ \leq \mu(D_1^c) + \mu(D_2^c) \end{array}$$

$$\mu(D_2^c) = 0.$$

$$\begin{aligned} 0 &\leq \mu(D^c) = \mu(D_1^c \cup D_2^c) \\ &\leq \underbrace{\mu(D_1^c)}_0 + \underbrace{\mu(D_2^c)}_0 \\ &= 0. \end{aligned}$$

can do it to combine
countably many a.e. statements.

note: $\left| \int f d\mu - \int g d\mu \right| = \left| \int (f-g) d\mu \right| \leq \int |f-g| d\mu.$

Pf: for any h , $h = h^+ - h^-$

$$\begin{aligned} \left| \int h d\mu \right| &= \left| \int h^+ d\mu - \int h^- d\mu \right| \\ &\leq \int h^+ d\mu + \int h^- d\mu \quad (\Delta\text{-ineq}) \\ &= \int (h^+ + h^-) d\mu = \int |h| d\mu. \end{aligned}$$

use $h = f - g$. (proved)

So $\left| \int f_n d\mu - \int f d\mu \right| \leq \int |f_n - f| d\mu$. (in DCT)

$E|X_n - X| \rightarrow 0 \Rightarrow E(X_n) \rightarrow E(X)$
but $\Leftarrow$ may not hold in general.

Recall: Cor. 2.3.6 $\nu(A) = \int_A f d\mu$, $f \geq 0$, $A \in \mathcal{F}$
is a measure

in this case, we have

$$\int h d\nu = \int h \cdot f d\mu \quad *$$

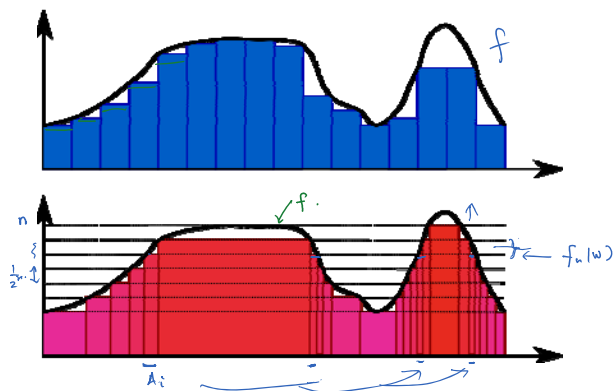
proof:

$$\nu(A) = \int 1_A f d\mu.$$

$$\text{So, } \int 1_A d\nu = \int 1_A f d\mu.$$

So statement (*) holds for $h = 1_A$.
 now prove (*) for h simple ≥ 0 , $h \geq 0$ ($h_n \uparrow h$)
 and general h .

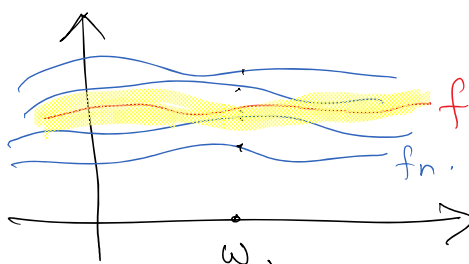
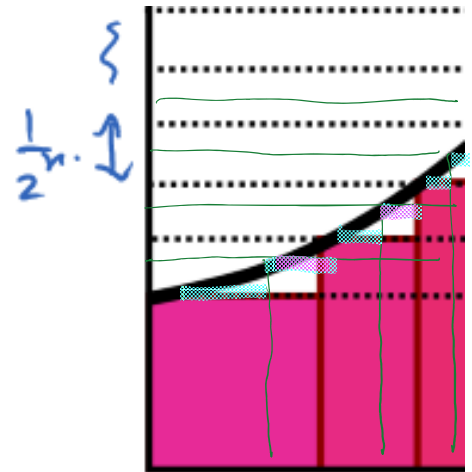
So in ch. 4, we will see if $\mu(A) = \int f d\mu$
 $\int h d\mu = \int h \cdot f d\mu = \oint h \cdot f \leftarrow \text{Rint.}$



<https://www.youtube.com/watch?v=AsMBB104EwE>

$$\left. \begin{array}{l} \text{fix } n \geq 1. \\ N_n = n 2^n \\ \delta_n = \frac{1}{2^n} \end{array} \right\}$$

$$\begin{array}{l} f_2 \geq f \\ f_{n+1} \geq f \\ f_n \uparrow f \end{array}$$



$$\begin{aligned} \mu(f > 0) &= \mu(\{\omega : f(\omega) > 0\}) \\ &= \mu(f^{-1}(0, \infty)) \end{aligned}$$

$P(X > 0)$ similar